Analysis of ATM/CRM Infrastructure in India (2023–2024) using Python: Insights and Recommendations for Digital Banking Adoption

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Project Overview:

This project analyses ATM/CRM infrastructure across different bank categories in India during 2023 and 2024. Using Python, the data is cleaned, explored, and analysed to identify infrastructure distribution, relationships, and year-wise trends. The findings highlight differences across bank categories and provide insights to support banking infrastructure planning and digital banking development.

Data Source:

- Dataset Name: Bankwise ATM POS Card Statistics 2022 onwards
- Source: India Data Portal (https://indiadataportal.com/p/reserve-bank-of-india/r/rbi-bankwise_atm_pos-in-mn-bx)
- File Type: CSV
- Selected analysis period: January 2023 - December 2024
- Number of Records: 1507
- Number of Attributes: 30

The dataset contains bank-level data on ATMs, PoS terminals, QR code adoption and transaction volumes and values for both credit and debit cards.

Problem Statement:

The challenge is to analyse bank-wise data to understand how ATM/CRM infrastructure differs across bank categories, examine the adoption of digital payment technologies over time.

Objectives:

- Analyse the distribution of on-site and off-site ATM/CRM infrastructure across different bank categories in India during 2023 and 2024.
- Perform statistical analysis of ATM/CRM counts to understand their distribution, central tendency, and variability.
- Examine the relationship between on-site and off-site ATM/CRM infrastructure using bivariate analysis and correlation.
- Compare ATM/CRM infrastructure across bank categories and identify year-wise changes between 2023 and 2024.
- Provide data-driven recommendations for leveraging existing ATM/CRM infrastructure and promoting digital banking technologies such as UPI, mobile banking, and online banking services.

Attribute Details:

S. No.	Attribute Name	Description
1	record_id	Unique identification number assigned to each record.
2	report_date	Reporting date for the bank-level statistics.
3	bank_category	Category of the bank, such as Public Sector, Private Sector, Foreign, Payment, or Small Finance Bank.
4	bank_name	Name of the bank associated with the reported statistics.
5	atms_crms_on_site	Number of ATMs and Cash Recycling Machines (CRMs) located on the bank's premises or within its branch premises.
6	atms_crms_off_site	Number of ATMs and CRMs located away from bank branches, such as in public or commercial locations.
7	pos_terminals	Number of Point-of-Sale (POS) terminals used to facilitate card-based payments.
8	micro_atm_count	Number of micro-ATM devices used to provide basic banking services.
9	bharat_qr_count	Number of Bharat QR payment codes associated with the bank.
10	upi_qr_count	Number of UPI QR codes associated with the bank.
11	credit_card_count	Number of credit cards reported for the bank.
12	debit_card_count	Number of debit cards reported for the bank.
13–30	Other transaction-related attributes	Remaining attributes capture credit-card and debit-card transaction statistics, including transaction volumes and values across the payment categories available in the dataset.

Tools & Technologies:

Programming Language

• Python 3.10+

Libraries

• **Pandas** – Data manipulation and analysis

• **NumPy** – Numerical operations

• **Matplotlib** – Data visualization

• **Seaborn** – Statistical visualization

Development Environment

• Google Colab

Project Workflow:

Data Collection

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Initial EDA

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Data Inspection

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Data Cleaning

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Data Validation

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Feature Engineering

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Final EDA

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Statistical Analysis

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Univariate Analysis

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Bivariate Analysis

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Multivariate Analysis

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Business Insights

Data Pre-Processing and Feature Engineering:

The dataset was prepared for analysis using data preprocessing techniques.

Step 1: Data Loading

- Imported the required Python libraries, including Pandas, NumPy, Matplotlib, and Seaborn.
- Loaded the Bankwise ATM POS Card Statistics dataset from GitHub into a Pandas DataFrame.
- Selected the analysis period from January 2023 to December 2024.

Step 2: Initial Data Inspection

- Checked the first few records using head().
- Identified the number of rows and columns using shape.
- Examined column names, data types, and non-null values using info().
- Generated descriptive statistics using describe().

Step 3: Handling Missing Values

- Identified missing values in the dataset.
- Found missing values in the atms_crms_onsite and atms_crms_offsite columns.
- Used median imputation to fill missing values.
- Selected the median because ATM/CRM data is right-skewed and contains potential outliers.

Step 4: Checking Duplicate Records

- Checked for duplicate rows to ensure data consistency.
- Confirmed that duplicate records were not present in the dataset.

Step 5: Renaming Column Names

- Renamed selected columns to improve readability and maintain consistent naming conventions.
- Used descriptive names for ATM/CRM, POS, QR code, and card-related attributes.

Step 6: Data Type Conversion

- Converted the reporting date column into the datetime format.
- Ensured that date values could be used for time-based analysis.

Step 7: Feature Engineering

- Created new features from the reporting date to support monthly, quarterly, and yearly analysis.

Feature	Description
year	Extracted the year from the reporting date.
month	Extracted the numerical month.
month_name	Extracted the month name.
quarter	Identified the reporting quarter.
bank_key	Combined bank category and bank name to create a bank identifier.

Step 8: Final Data Validation

- Verified the final dataset dimensions.
- Checked missing values after imputation.

- Rechecked duplicate records.
- Confirmed data types and reporting date range.
- Reviewed numerical and categorical summaries.

Step 9: Prepared Dataset for Exploratory Data Analysis

- Prepared the cleaned dataset for statistical analysis and visualization.
- Used the transformed data to analyse ATM/CRM infrastructure across bank categories.
- Prepared date-based features for examining yearly and monthly trends.
- Identified digital payment attributes such as UPI QR, Bharat QR, POS terminals, and micro-ATMs for further analysis.

Statistical Analysis:

Descriptive statistical techniques were used to understand the central tendency, variability, distribution shape, and extreme observations in the major numerical variables. The techniques include Mean, Median, Mode, Variance, Standard Deviation, Skewness, and Kurtosis.

	Mean	Median	Mode	Variance	Standard Deviation
record_id	1342	1342	589	1.89E+05	4.35E+02
atms_crms_on_site	2004.81	449	0	1.60E+07	4.00E+03
atms_crms_off_site	1468.97	35	0	2.67E+07	5.17E+03
pos_terminals	137234.3	1413	0	1.62E+11	4.03E+05
micro_atm_count	24689.05	141	0	6.23E+09	7.89E+04
bharat_qr_count	94730.2	0	0	8.95E+10	2.99E+05
upi_qr_count	6443078	45909	0	1.61E+15	4.01E+07
credit_card_count	1543920	10863	0	1.79E+13	4.23E+06
debit_card_count	15416074	3791835	0	1.19E+15	3.45E+07
cc_pay_trns_at_pos_vol	2705693	26940	0	7.24E+13	8.51E+06
cc_pay_trns_at_pos_val	9212808	163486.3	0	7.20E+14	2.68E+07

cc_pay_trns_online_vol	2472392	9725	0	6.25E+13	7.90E+06
cc_pay_trns_online_val	15811485	72430.83	0	2.35E+15	4.85E+07
cc_pay_trns_other_vol	421.69	0	0	1.42E+08	1.19E+04
cc_pay_trns_other_val	130.45	0	0	4.16E+06	2.04E+03
cc_cash_withdraw_atm_val	12924.14	0	0	1.15E+09	3.40E+04
cc_cash_withdraw_atm_vol	64264.17	0	0	3.40E+10	1.84E+05
dc_pay_trns_at_pos_vol	2045449	250083	0	3.03E+13	5.50E+06
dc_pay_trns_at_pos_val	5025126	565569.4	0	1.76E+14	1.33E+07
dc_pay_trns_online_vol	781503.8	98887	0	4.31E+12	2.08E+06
dc_pay_trns_online_val	2471792	254855.2	0	4.33E+13	6.58E+06
dc_pay_trns_other_vol	2224.93	0	0	1.21E+08	1.10E+04
dc_pay_trns_other_val	35969.53	0	0	2.90E+10	1.70E+05
dc_cash_withdraw_atm_val	8538423	1370925	0	5.44E+14	2.33E+07
dc_cash_withdraw_atm_vol	42193589	6641232	0	1.40E+16	1.18E+08
dc_cash_withdraw_pos_val	1635.71	0	0	5.44E+07	7.38E+03

dc_cash_withdraw_pos_vol	1613.61	0	0	5.27E+07	7.26E+03
year	2023.51	2024	2024	2.50E-01	5.00E-01
month	6.51	7	12	1.20E+01	3.46E+00

Data Visualization:

Visualization techniques were used to represent data graphically and make patterns easier to understand.

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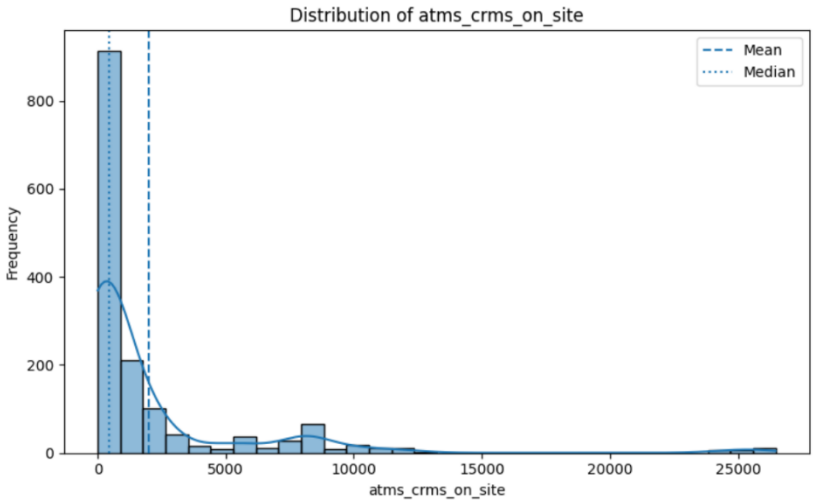
1. Univariate Analysis

The **atms_crms_on_site** column was selected as an example because it represents on-site ATM/CRM infrastructure. Analysing it helps us understand:

- The average number of on-site ATM/CRM facilities.
- How infrastructure differs across bank categories.
- Whether the distribution is skewed.
- How infrastructure changed between 2023 and 2024.
- Whether there are unusually high or low observations.

Analysis Performed:

1. **Histogram - Mean vs Media**



Interpretation:

1. The median is approximately 400, meaning around half of the observations have on-site ATM/CRM counts below this value and half above it.

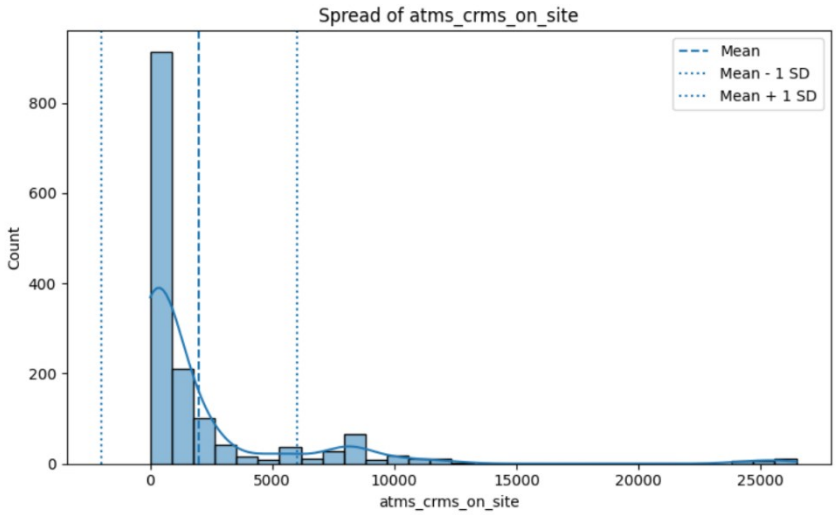
2. The mean is approximately 1,900, which is much higher than the median.

3. Most observations are concentrated below approximately 2,000, while a few observations extend beyond 10,000.

4. Some observations reach approximately 26,000, contributing to the long right tail.

5. The difference between the mean and median indicates that the distribution is influenced by a small number of very high values. Hence, the distribution is highly right-skewed (positively skewed).

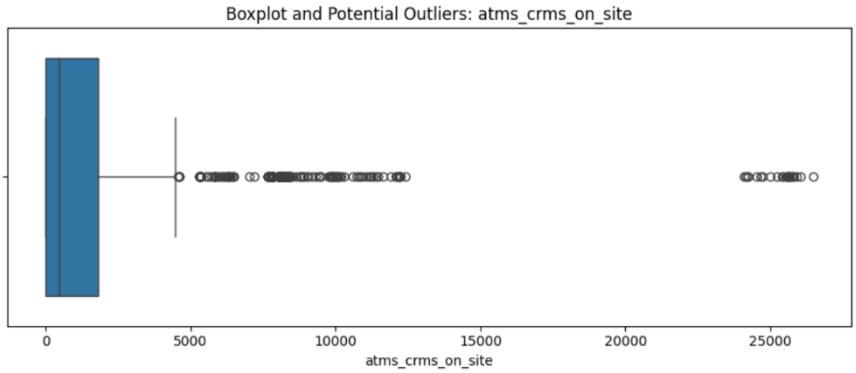
2. Histogram - Standard Deviation



Interpretation:

1. The mean on-site ATM/CRM count is approximately 1,900.
2. The standard deviation is approximately 3,700, indicating a wide spread of values around the mean.

3. Boxplot and Potential Outliers



Interpretation:

Median: The line inside the box represents the median number of on-site ATMs/CRMs.

Interquartile range (IQR): The box represents the middle 50% of observations, between the 25th percentile (Q1) and the 75th percentile (Q3).

Whiskers: The whiskers show the range of values within the standard (1.5 \times IQR) outlier limits.

Potential outliers: The circles beyond the upper whisker represent observations with unusually high ATM/CRM counts. Some values are around 5,000–12,500, while another group lies around 24,000–26,500.

The boxplot of atms_crms_on_site reveals a **strongly right-skewed distribution** with several potential upper-end outliers. Most observations have relatively low on-site ATM/CRM counts, while a small number have substantially higher counts. These extreme values may reflect differences in banking infrastructure across banks rather than data-entry errors.

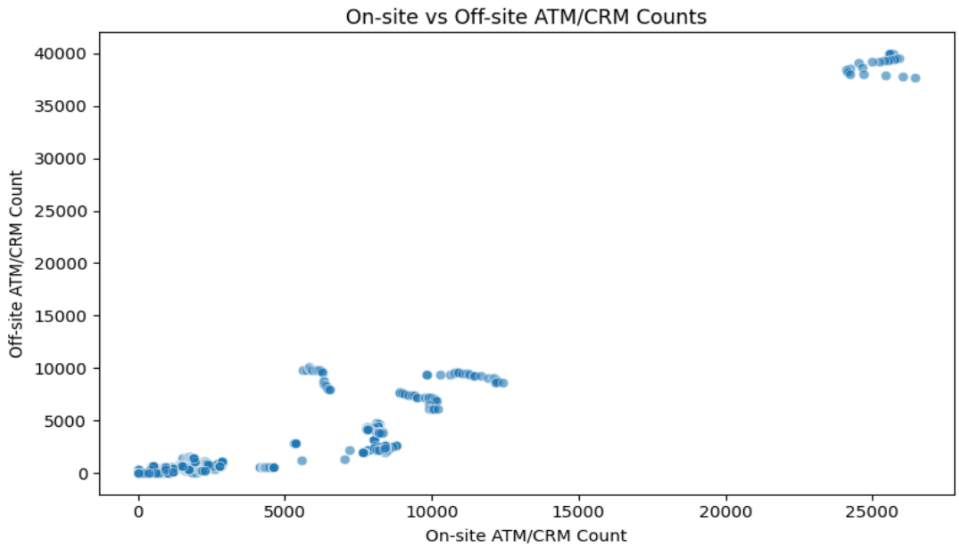
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2. Bivariate Analysis

Bivariate analysis was conducted to examine the relationship between on-site and off-site ATM/CRM infrastructure. A scatter plot and Pearson correlation coefficient were used to understand the direction and strength of the association between the two variables.

Analysis Performed:

1. **Scatterplot — on-site vs. off-site ATM/CRM counts**



Interpretation:

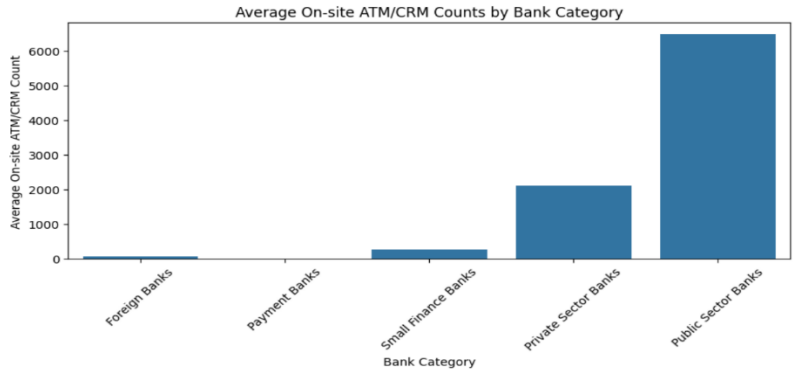
1. The plot generally shows a positive relationship: observations with higher on-site counts tend to have higher off-site counts.

2. The observations are not evenly distributed. Several visible clusters appear, suggesting differences in infrastructure scale across banks or bank categories.

3. The upper-right cluster contains observations with approximately 24,000–26,000 on-site and 37,500–40,000 off-site ATM/CRM counts.

4. Some observations between approximately 5,000 and 7,000 on-site have off-site counts around 8,000–10,000, showing that off-site infrastructure can exceed on-site infrastructure.

2. Barplot — average on-site ATM/CRM counts by bank category



Interpretation:

1. Public Sector Banks recorded the highest average, at approximately 6,500 on-site ATM/CRM facilities.

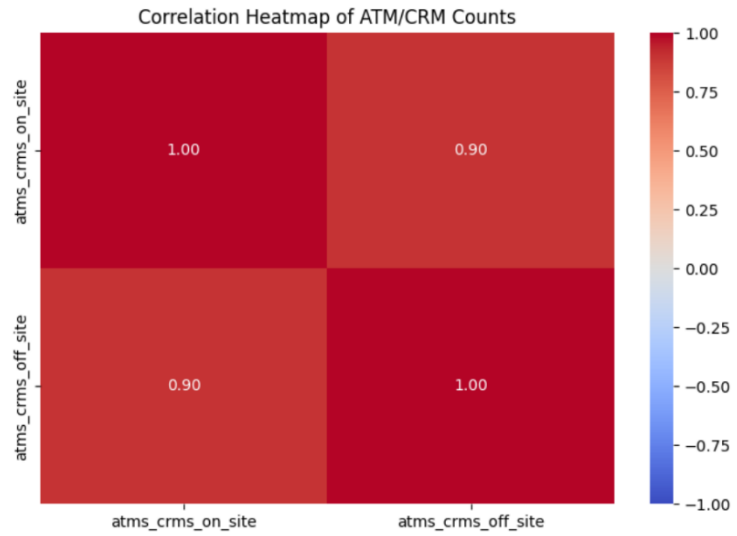
2. Private Sector Banks ranked second, with an average of approximately 2,100.

3. Small Finance Banks had a much lower average of approximately 270.

4. Foreign Banks recorded approximately 70 on-site ATM/CRM facilities on average.

5. Payment Banks had an average close to zero, indicating very limited on-site ATM/CRM infrastructure in this dataset.

3. Correlation heatmap — numerical variables



Interpretation:

1. The correlation between on-site and off-site ATM/CRM counts is +0.90.

2. A correlation value of +0.90 is close to +1, indicating a strong positive linear relationship.

3. This means that observations with higher on-site ATM/CRM counts generally tend to have higher off-site ATM/CRM counts.

4. The diagonal values are 1.00 because each variable is perfectly correlated with itself.

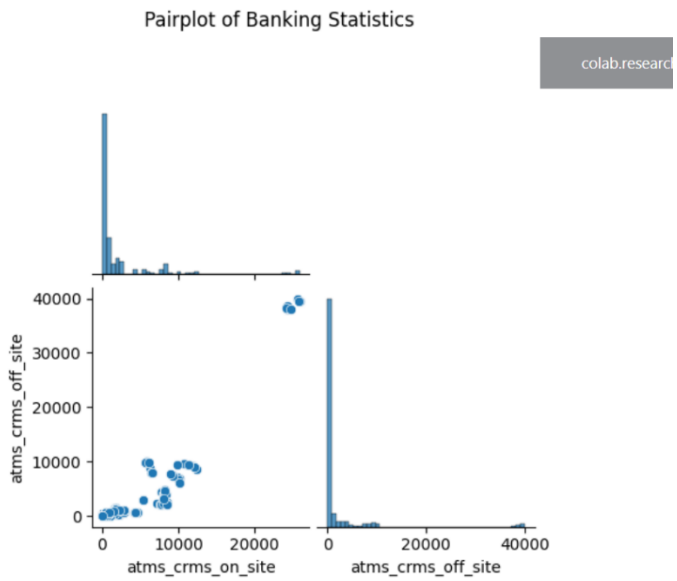
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3. Multivariate Analysis

Multivariate analysis was used to analyze relationships among multiple variables.

Analysis Performed:

1. **Pairplot – relationships among numerical variables**



Interpretation:

Distribution: Both variables are highly right-skewed. Most observations have relatively low counts, while a few observations have very high values.

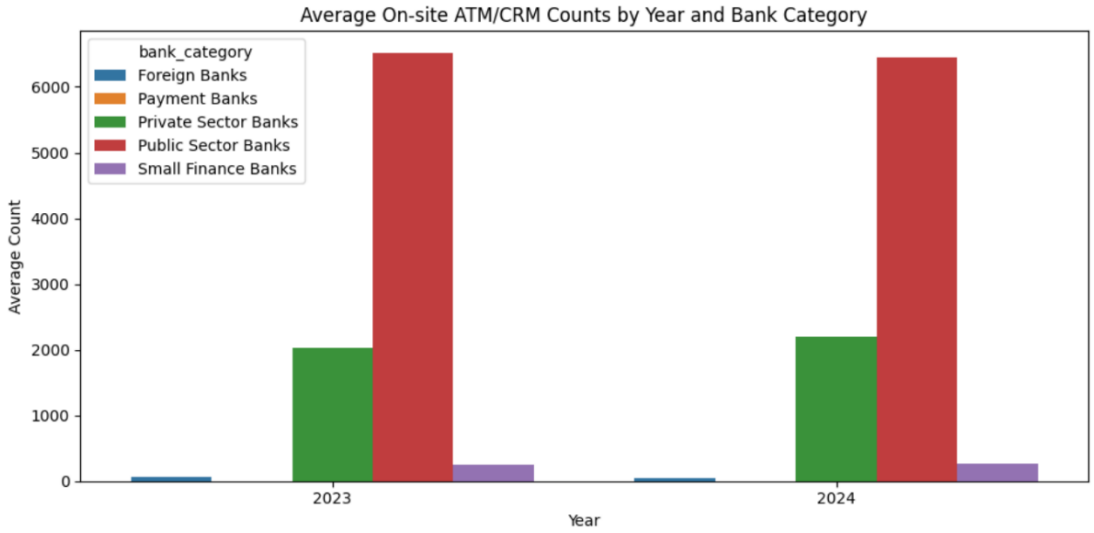
On-site ATM/CRM: Most observations are concentrated below approximately 3,000, with a few reaching around 26,000.

Off-site ATM/CRM: Most observations are concentrated below approximately 1,500, while some reach approximately 40,000.

Relationship: The scatter plot shows a strong positive association. The correlation coefficient of approximately +0.90 indicates that higher on-site counts generally correspond to higher off-site counts.

Clusters: The plot contains distinct groups of observations, suggesting differences in infrastructure scale across banks or banking categories.

2. Grouped analysis — bank category and reporting year



Interpretation:

1. Public Sector Banks had the highest average on-site ATM/CRM infrastructure in both years. Their average decreased slightly from approximately 6,523 in 2023 to 6,454 in 2024, a decline of around 1.05%.

2. Private Sector Banks showed an increase from approximately 2,029 to 2,198, representing growth of around 8.36%.

3. Small Finance Banks increased from approximately 243 to 274, showing growth of around 12.71%.

4. Foreign Banks decreased from approximately 66 to 43, a decline of around 34.87%.

5. Payment Banks decreased from approximately 0.17 to 0.03. Although the percentage decline is large, the absolute infrastructure count was already very low.

3. Pivot table — bank category, year, and ATM counts

Year	**2023**	**2024**
bank_category		
Foreign Banks	66.18	43.10
Payment Banks	0.17	0.03
Private Sector Banks	2028.55	2198.10
Public Sector Banks	6522.85	6454.44

Year	2023	2024
bank_category		
Small Finance Banks	242.90	273.79

Summary of Findings:

1. Public Sector Banks have the largest ATM/CRM infrastructure.

They recorded the highest average counts in both years, decreasing slightly from 6,522.85 in 2023 to 6,454.44 in 2024. This indicates that public sector banks maintain a comparatively extensive physical banking network.

2. Private Sector Banks showed growth in on-site infrastructure.

The average on-site ATM/CRM count increased from approximately 2,029 in 2023 to 2,198 in 2024, representing growth of around 8.36%.

3. Small Finance Banks recorded infrastructure growth.

Their average on-site ATM/CRM count increased from approximately 243 in 2023 to 274 in 2024, a rise of around 12.71%.

4. On-site and off-site ATM/CRM counts have a strong positive relationship.

The correlation is approximately +0.90, suggesting that banks with more on-site ATMs/CRMs also tend to have more off-site ATMs/CRMs. This indicates an association, not causation.

5. Infrastructure is unevenly distributed across bank categories.

Public and private sector banks account for much higher average ATM/CRM counts than foreign and payment banks. Payment banks have very limited ATM/CRM infrastructure in comparison.

6. The findings can support digital banking recommendations.

Banks may use their existing ATM/CRM networks to improve customer awareness and access to digital services such as UPI, mobile banking, and internet banking.

Types of Analysis:

1. Descriptive Analysis

- Examined the mean, median, standard deviation, skewness, and kurtosis of numerical variables.
- Analysed the distribution of on-site and off-site ATM/CRM counts using histograms and boxplots.
- Compared average on-site ATM/CRM counts across bank categories.
- Used grouped analysis and pivot tables to compare 2023 and 2024.
- Observed that Public Sector Banks had the highest average on-site ATM/CRM counts in both years.

Finding: Banking infrastructure varies considerably across categories. Private Sector Banks and Small Finance Banks showed increases in average on-site ATM/CRM counts, while Public Sector Banks and Foreign Banks showed decreases.

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2. Diagnostic Analysis

- Used scatterplots to examine the relationship between on-site and off-site ATM/CRM counts.
- Applied correlation analysis to measure the strength of the relationship.
- Used boxplots to identify potential outliers.
- Compared infrastructure averages across bank categories and reporting years.

Finding: The strong positive correlation suggests that banks with larger on-site ATM/CRM networks generally tend to have larger off-site networks. Differences in bank category and infrastructure scale may contribute to the variation.

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3. Predictive Analysis

- Examined monthly ATM/CRM infrastructure trends.
- Compared historical monthly values across 2023 and 2024.
- Forecasting models can be proposed for selected ATM/CRM infrastructure measures.

Expected Outcome: Forecasting may provide an estimate of future ATM/CRM infrastructure trends and support preliminary infrastructure planning.

Limitation: The current analysis does not include validated forecasting results. Since the dataset covers only two years, predictions may be sensitive to short-term changes and should be interpreted cautiously.

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4. Prescriptive Analysis

- Identified bank categories showing increases or decreases in ATM/CRM infrastructure.
- Recommended evaluating infrastructure allocation alongside transaction activity and customer demand.
- Suggested examining UPI QR and POS adoption to identify potential opportunities for digital payment expansion, where relevant data is available.
- Recommended analysing credit- and debit-card transaction volumes and values to understand payment activity, where relevant data is available.
- Suggested considering demand, usage, operating costs, and geographic coverage when planning infrastructure.

Expected Outcome: The findings can support better-informed decisions regarding ATM/CRM infrastructure planning and the promotion of digital banking services.

Limitation: The current dataset alone cannot confirm where infrastructure expansion or reduction is necessary. Additional information on customer demand, transaction activity, operating costs, and geographic coverage would be required for stronger recommendations.

Future Enhancements and Suggestions:

Based on these findings, I would recommend

Public Sector Banks: Use their extensive ATM/CRM networks to conduct digital payment awareness campaigns, especially for customers who may need assistance using UPI or mobile banking.

Private Sector Banks: Explore integrating digital banking promotions and QR-payment awareness into their existing ATM/CRM customer experience.

Banks with smaller ATM/CRM networks: Consider promoting digital-first services such as UPI and mobile banking to reduce dependence on physical banking infrastructure.

This would help banks identify opportunities to improve digital payment adoption while making better use of their existing banking infrastructure.

Conclusion:

The analysis highlights significant differences in on-site ATM/CRM infrastructure across Indian bank categories during 2023–2024. Public Sector Banks maintained the highest average infrastructure counts, while Private Sector Banks and Small Finance Banks showed increases in their averages. A strong positive relationship was also observed between on-site and off-site ATM/CRM counts.
